

Customer Churn Analysis

1. Project Overview

Analyzed customer data to identify key factors contributing to customer churn and retention. Evaluated customer demographics, tenure, spending behavior, subscription plans, payment methods, and service interactions to identify high-risk customer segments. Developed actionable insights and recommendations to help reduce churn, improve customer retention, and minimize potential revenue loss.

2. Dataset Summary

- Rows: 542 - Columns: 18 - Key Features:
- Customer demographics (Gender, Age, State, City)
- Subscription details (Tenure_Months, Subscription_Type, Monthly_Charges , Total_Charges Contract_Type, , Internet_Service)
- Churn Customer (Payment_Method , Tech_Support, Senior_Citizen, Dependents, Churn, Last_Interaction_Date)

3. Exploratory Data Analysis using Python

Python and Pandas were used to clean and prepare the dataset.

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

	Customer_ID	Customer_Name	Gender	Age	State	City	Tenure_Months	Subscription_Type	Monthly_Charges	Total_Charges	Contract_Type
0	C1000	Abhishek Jaiswal	Female	35.0	Maharashtra	Pune	58.0	basic	1614.26	93273.08	One Year
1	C1001	Rohit Kumar	Male	NaN	Delhi	Noida	36.0	Premium	429.60	15858.60	Two Year
2	C1002	Pooja Singh	Female	42.0	UP	Kanpur	13.0	Basic	555.03	6869.39	Month-to-Month
3	C1003	NaN	Male	59.0	Karnataka	Bengaluru	55.0	Standard	NaN	101349.10	Two Year
4	C1004	Mohit Arora	male	26.0	Gujarat	Ahmedabad	6.0	Premium	314.02	1926.12	One Year

- **Missing Data Handling:** Check for null values and fill missing values with appropriate Values.

- **Data Consistency Check:** Data consistency checks like Age validation, Duplicate customer IDs and Charges consistency were performed to ensure that the values were logically compatible.
- **Column Standardization:** Convert column's values in proper case, convert numeric values, remove extra spaces, corrected value count and corrected Date datatype for better readability and documentation.
- **Feature Engineering:**
 - Created **Tenure_Group** column by binning tenure months.
 - Created **Customer_Value** column from purchase data.
 - Created **Senior_Flag** column from monthly charges and tenure data.
 - Created **Churn_Flag** column from churn status.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

MySQL was used to answer business questions and calculate important metrics.

1. Churn Rate

	Churn_Rate
▶	23.35

2. Churn by contract type

	Contract_Type	Churn_by_Contract_Type
►	One Year	152
	Month-To-Month	170
	Two Year	149

3. Churn by Internet Service

	Internet_Service	Churn_by_Internet_Service
►	Fiber	127
	5G	114
	Cable	115
	Dsl	115

4. Churn by State

	State	Churn_by_State
►	Gujarat	106
	Karnataka	98
	Up	94
	Delhi	93
	Maharashtra	80

5. Payment Method Wise Customers

	Payment_Method	Payment_Method_Wise_Customers
▶	Upi	97
	Debit Card	92
	Credit Card	92
	Cash	89
	Net Banking	87
	Unknown	14

6. Subscription Type Wise Customers

	Subscription_Type	Subscription_Type_Wise_Customers
▶	Premium	163
	Standard	158
	Basic	150

7. Highest Revenue State

	State	State_Revenue
▶	Gujarat	5162488.22
	Delhi	5068454.14
	Up	4724267.73
	Karnataka	4570115.7
	Maharashtra	4241775.94

8. Average Charges by Contract

	Contract_Type	Avg_Monthly_Charges
▶	One Year	1274.8449999999993
	Month-To-Month	1432.0493529411774
	Two Year	1356.9775167785224

9. Senior Citizen Churn

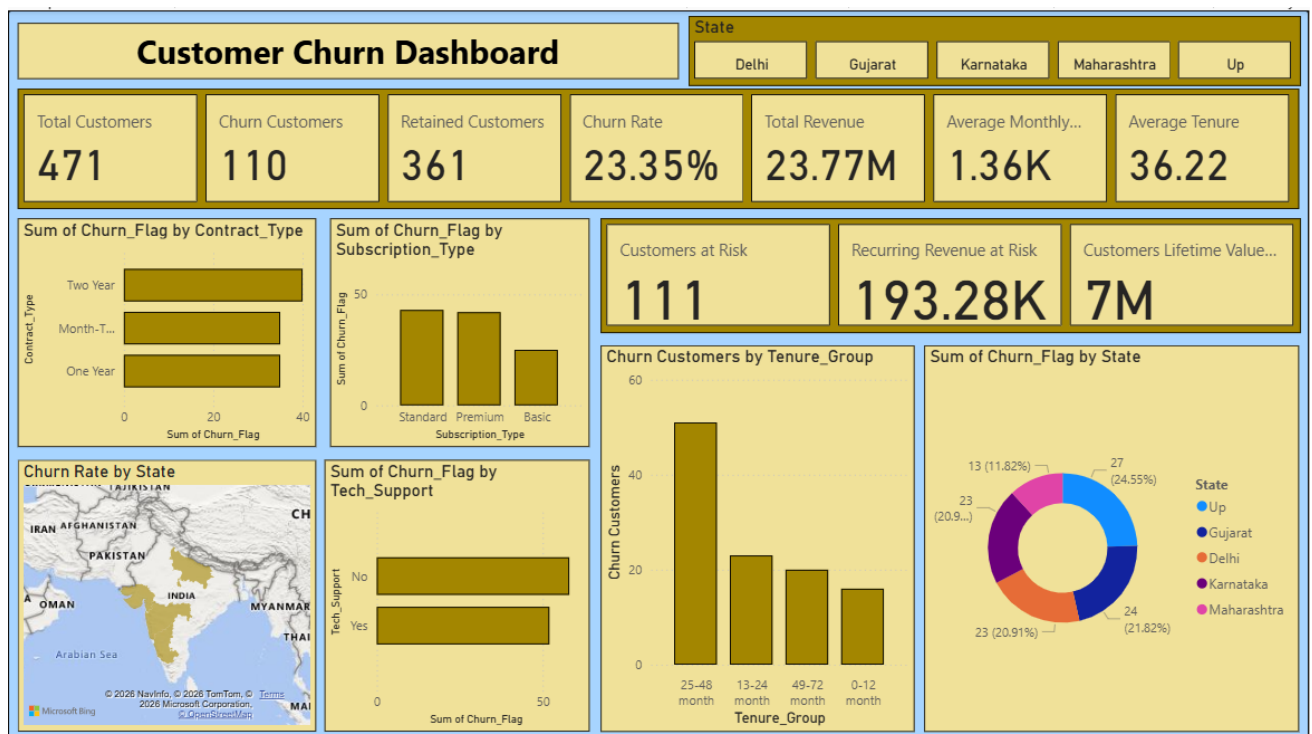
	Senior_Citizen	Senior_Citizen_Churn
►	No	90
	Yes	20

10. Revenue by Gender

	Gender	Revenue_by_Gender
►	Female	12697462.559999999
	Male	11069639.169999998

5. Churn Risk Analysis and Dashboard in Power BI

- **Churn Risk Analysis:** Created churn risk scores column and churn risk level column to calculate high risk customers and risk revenue.
- Built an interactive dashboard: Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Important Insights

Subscription Type: Some subscription plans showed higher churn rates than others, suggesting that plan structure and customer expectations may influence retention.

Tech Support: Customers with No Tech Support showed increased churn, indicating that unresolved customer issues may contribute to churn.

Risk of Churn: Customers at risk should be monitored to determine whether price sensitivity is associated with churn.

Tenure: Customers with **25-48 month** tenure showed a higher tendency to churn, need to know the reason behind it as it indicating dissatisfaction of customers.

State: State which has highest number of churn customers is UP.

7. Business Recommendations

Based on the analysis, the company could:

- 1. Improve customer support:** Customers with repeated complaints or support interactions should receive faster resolution.
- 2. Review pricing and plans:** Investigate whether high-paying customers perceive insufficient value.
- 3. Create targeted retention campaigns:** Offer suitable incentives to high-risk customers rather than giving discounts to everyone.
- 4. Monitor high-risk segments:** Use the churn-risk analysis to prioritize customers requiring intervention.
- 5. Track churn continuously:** Create a regular churn monitoring dashboard so management can identify changes quickly.